AQ4.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
The dimension of a vector space is the :
- [ ] Cardinality of any spanning set.
- [ ] Cardinality of a basis.
- [ ] Cardinality of minimal spanning set.
- [ ] Cardinality of maximal linearly independent set.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Cardinality of a basis.
Cardinality of minimal spanning set.
Cardinality of maximal linearly independent set.

1 point
Match the sets of vectors in column A with their properties of linear dependence or independence in column B and the dimension of the vector spaces in column C spanned by the sets.

- [ ] a → → ii → → 3, b → → iii → → 2
- [ ] c → → iii → → 2, d → → iv → → 1
- [ ] a → → ii → → 4, b → → i → → 3
- [ ] c → → iv → → 1, d → → iii → → 2

No, the answer is incorrect.
Score: 0
Accepted Answers:
a → → ii → → 4, b → → i → → 3
c → → iv → → 1, d → → iii → → 2

1 point
Choose the set of correct options.
- [ ] The dimension of the vector space $M_{1\times 2}(R)$M1×2(R) is 2.
- [ ] The dimension of the vector space $M_{2\times 1}(R)$M2×1(R) is 1.
- [ ] The dimension of the vector space $M_{3\times 3}(R)$M3×3(R) is 3.
- [ ] A basis of $M_{2\times 2}(R)$M2×2(R) is the set $\{\begin{bmatrix}1&0\\0&0\end{bmatrix}$\{[1000], $\begin{bmatrix}0&1\\0&0\end{bmatrix}$[0010], $\begin{bmatrix}0&0\\1&0\end{bmatrix}$[0100], $\begin{bmatrix}0&0\\0&1\end{bmatrix}\}$[0001]\}

No, the answer is incorrect.
Score: 0
Accepted Answers:
The dimension of the vector space $M_{1\times 2}(R)$M1×2(R) is 2.

A basis of $M_{2\times 2}(R)$ is the set $\left\{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&1\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}, \begin{bmatrix}0&0\\0&1\end{bmatrix}\right\}$

1 point

Consider the following sets:

- $V_1 = \{A \mid A \in M_{2\times 2}(R)$ and A is a symmetric matrix, i.e., $A = A^T\}$
- $V_2 = \{A \mid A \in M_{2\times 2}(R)$ and A is a scalar matrix$\}$
- $V_3 = \{A \mid A \in M_{2\times 2}(R)$ and A is a diagonal matrix$\}$
- $V_4 = \{A \mid A \in M_{2\times 2}(R)$ and A is an upper triangular matrix$\}$
- $V_5 = \{A \mid A \in M_{2\times 2}(R)$ and A is a lower triangular matrix$\}$

All $V_i, i = 1, 2, 3, 4, 5$ are subspaces of the vector space $M_{2\times 2}(R)$.

Choose the set of correct options.

- [ ] The dimension of V_1 is 3.
- [ ] The dimension of V_2 is 3.
- [ ] The dimension of V_3 is 1.
- [ ] The dimension of V_4 is 3.
- [ ] The dimension of V_5 is 3.

No, the answer is incorrect.
Score: 0
Accepted Answers:

The dimension of V_1 is 3.
The dimension of V_4 is 3.
The dimension of V_5 is 3.

1 point

Consider the following two matrices M_1 and M_2.

$$M_1 = \begin{bmatrix}1&4&3&2\\2&3&1&4\\3&4&2&1\end{bmatrix}, M_2 = \begin{bmatrix}1&0&0&0\\1&0&1&0\\0&1&0&1\end{bmatrix}$$

Choose the correct set of options.

- [ ] The row rank of both the matrices, M_1 and M_2, is 2.
- [ ] The row rank of both the matrices, M_1 and M_2, is 3.
- [ ] The column rank of M_1 is 4, but the column rank of M_2 is 3.
- [ ] The column rank of M_2 is 4, but the column rank of M_1 is 3.
- [ ]

The column rank of both the matrices, M_1 M1 and M_2 M2, is 4.

☐

The column rank of both the matrices, M_1 M1 and M_2 M2, is 3.

No, the answer is incorrect.
Score: 0
Accepted Answers:

The row rank of both the matrices, M_1 M1 and M_2 M2, is 3.
The column rank of both the matrices, M_1 M1 and M_2 M2, is 3.

Level 2

Find the dimension of the vector space
$V = \{A \mid \text{ sum of entries in each row is 0, and } A \in M_{3\times 2}(R)\}$.V =
{A ∣ sum of entries in each row is 0, and A ∈ M3×2(R)}.

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 3

1 point

Find the dimension of the vector space
$V = \{(x, y, z, w) \mid x + y = z + w, x + w = y + z, \text{ and } x, y, z, w \in R\}$.V =
{(x, y, z, w) ∣ x + y = z + w, x + w = y + z, and x, y, z, w ∈ R}.

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 2

1 point

1 point

Choose the set of correct options.

☐

If $S \subset R^2$ S ⊂ R2 contains three vectors, then vectors in SS must be linearly independent.

☐

If $S \subset R^3$ S ⊂ R3 contains three vectors, then vectors in SS may or may not be linearly independent.

☐

If $S \subset R^4$ S ⊂ R4 contains five vectors, then vectors in SS must be linearly dependent.

☐

If $S \subset R$S ⊂ R contains one vector, then the vector in SS must be linearly dependent.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If $S \subset R^3$ S ⊂ R3 contains three vectors, then vectors in SS may or may not be linearly independent.
If $S \subset R^4$ S ⊂ R4 contains five vectors, then vectors in SS must be linearly dependent.

Find the dimension of the vector space $V = \{A \in M_{3\times 3}(R) : A = A^T\}$V = {A ∈ M3×3
(R) : A = AT }.

No, the answer is incorrect.
Score: 0
Accepted Answers:

(Type: Numeric) 6

1 point

Find the dimension of the vector space $V = \{A \in M_{3\times 3}(R): A = -A^T\}$V = {A ∈ M3×3 (R) : A = −AT }.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Check Answers

Your score is: 0/10